

Chain-of-Thought (CoT): Asking the AI to "Think Step-by-Step"

Spoken Tutorial

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Welcome

Learning Objective

In this tutorial, we will learn,

- ▶ **By the end of this tutorial, you will be able to use Chain-of-Thought prompting to get better responses from AI models.**



System Requirements

- ▶ **To follow this tutorial, you need a web browser like Chrome or Firefox. You also need an active...**



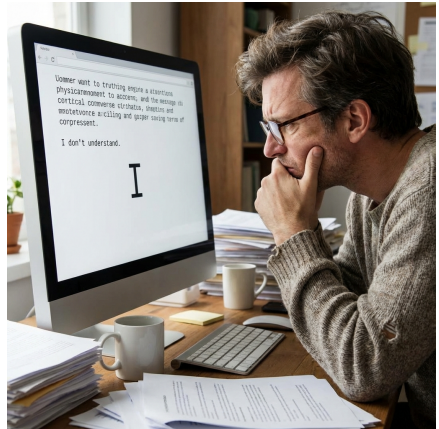
Prerequisites

- ▶ **Basic understanding of AI chatbots and how to write simple prompts**



Have you ever got a short or incomplete answer from an AI?

- ▶ **Got short or confusing AI answers?**
- ▶ **One-line response, need detail**
- ▶ **This is a common AI**



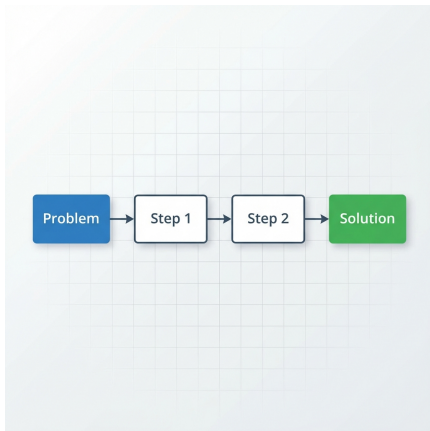
Introducing Chain-of-Thought (CoT)

- ▶ **AI lacks step-by-step thinking**
- ▶ **CoT prompts AI to 'think' before answering**



What is Chain-of-Thought?

- ▶ **AI breaks down problems**
- ▶ **AI explains solution steps**
- ▶ **Show steps, not just final answer**



Let's try an example (Bad Prompt)



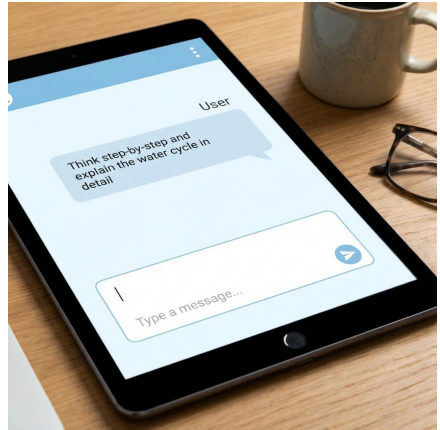
Bad Prompt - Result

- ▶ **AI gives very short answers**
- ▶ **Answer correct, but not detailed**

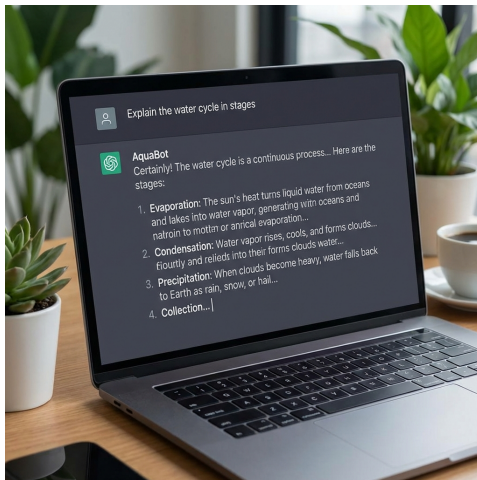


Now, let's use CoT (Good Prompt)

- ▶ Add "Think step-by-step" to prompt
- ▶ Ask AI to explain water cycle
- ▶ Submit prompt in chatbot

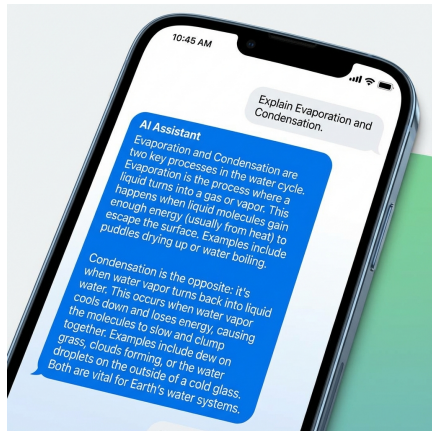


Good Prompt - Result (Part 1)



Good Prompt - Result (Part 2)

- ▶ **AI clearly explains stages**
- ▶ **Provides specific examples**
- ▶ **Easy-to-understand explanations**



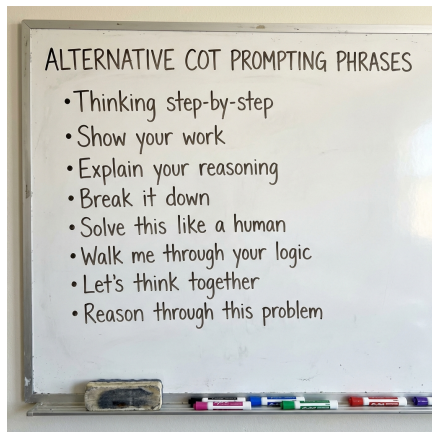
Why CoT works

- ▶ **AI processes information logically**
- ▶ **Breaks down tasks into smaller steps**
- ▶ **Achieves better quality answers**



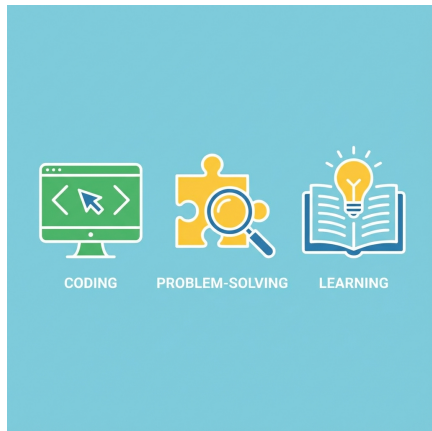
Other ways to ask for CoT

- ▶ Use alternative CoT phrases
- ▶ Try 'Walk me through your reasoning'
- ▶ Try 'Explain your thought process'
- ▶ Try 'Break this



When to use CoT

- ▶ **For detailed explanations**
- ▶ **For logical thinking**
- ▶ **Aids coding and finding errors**
- ▶ **Clarifies difficult topics**



Summary

In this tutorial, we learned that:

Assignment

- **For your assignment, open an AI chatbot. First, ask it: 'Explain how a car engine works' without...**

Domain Expert
(Domain Expert)
Affiliation

Thank you